Calculus Quiz: Convergence and Divergence of Sequences

Practice Test & Evaluation Worksheet

Student Name: _____ **Date:** _____

Score: _____ / 10 **Topic:** Infinite Sequences & Limits

Instructions: For each of the following sequences a_n (where $n = 1, 2, 3, \dots$), evaluate the limit as $n \to \infty$ and determine whether the sequence is **convergent**, **divergent**, or **oscillatory**. Select the correct option for each question.

Q1. Consider the sequence defined by terms:

$$\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \dots,\ \frac{n}{n+1},\ \dots$$

Determine the behavior and limit of $a_n = \dfrac{n}{n+1}$ as $n \to \infty$.

(A) Convergent with limit $= 1$

(B) Convergent with limit $= 0$

(C) Divergent (approaches $+\infty$)

(D) Oscillatory

Q2. Evaluate the long-term behavior of the sequence $a_n = \dfrac{2^n - 1}{2^n}$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= 1$

(C) Divergent to $+\infty$

(D) Oscillatory between 0 and 1

Q3. Consider the sequence $a_n = \dfrac{2^n - 1}{3^n}$. What is the limit of this sequence as $n \to \infty$?

(A) Convergent with limit $= 0$

(B) Convergent with limit $= \dfrac{2}{3}$

(C) Convergent with limit $= 1$

(D) Divergent to $+\infty$

Q4. Find the limit of the trigonometric sequence $a_n = n \sin\left(\dfrac{1}{n}\right)$ as $n \to \infty$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= 1$

(C) Convergent with limit $= \pi$

(D) Divergent / Undefined

Q5. Determine the convergence or divergence of the rational sequence $a_n = \dfrac{n+3}{n^2+5n+6}$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= 1$

(C) Convergent with limit $= \dfrac{1}{2}$

(D) Divergent to $+\infty$

Q6. Using order of magnitude comparison or L'Hôpital's Rule, find the limit of $a_n = \dfrac{n}{2^n}$ as $n \to \infty$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= 1$

(C) Convergent with limit $= \dfrac{1}{2}$

(D) Divergent to $+\infty$

Q7. Evaluate the limit of the sequence $a_n = \left(\dfrac{n}{n+1}\right)^n$ as $n \to \infty$.

(A) Convergent with limit $= e$

(B) Convergent with limit $= \dfrac{1}{e}$

(C) Convergent with limit $= 1$

(D) Divergent

Q8. Determine the limit of the algebraic radical sequence $a_n = \dfrac{1}{\sqrt{n^2-1} - \sqrt{n^2+n}}$ as $n \to \infty$.

(A) Convergent with limit $= -2$

(B) Convergent with limit $= 2$

(C) Convergent with limit $= 0$

(D) Divergent to $-\infty$

Q9. Find the limit of the sequence $a_n = n - \sqrt{n^2-n}$ as $n \to \infty$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= \dfrac{1}{2}$

(C) Convergent with limit $= 1$

(D) Divergent to $+\infty$

Q10. Analyze the behavior of the geometric sequence $a_n = \dfrac{1}{(0.9)^n}$ as $n \to \infty$.

(A) Convergent with limit $= 0$

(B) Convergent with limit $= 0.9$

(C) Divergent (approaches $+\infty$)

(D) Oscillatory

Answer Key & Step-by-Step Solutions

Complete Explanations for Calculus Quiz

Question	Correct Answer	Limit Value / Summary
Q1	**A**	Convergent, $\lim_{n\to\infty} a_n = 1$
Q2	**B**	Convergent, $\lim_{n\to\infty} a_n = 1$
Q3	**A**	Convergent, $\lim_{n\to\infty} a_n = 0$
Q4	**B**	Convergent, $\lim_{n\to\infty} a_n = 1$
Q5	**A**	Convergent, $\lim_{n\to\infty} a_n = 0$
Q6	**A**	Convergent, $\lim_{n\to\infty} a_n = 0$
Q7	**B**	Convergent, $\lim_{n\to\infty} a_n = 1/e$
Q8	**A**	Convergent, $\lim_{n\to\infty} a_n = -2$
Q9	**B**	Convergent, $\lim_{n\to\infty} a_n = 1/2$
Q10	**C**	Divergent, $\lim_{n\to\infty} a_n = +\infty$

Detailed Solutions

Q1 Solution (A): $\lim_{n\to\infty} \dfrac{n}{n+1} = \lim_{n\to\infty} \dfrac{1}{1+1/n} = \dfrac{1}{1+0} = 1$. The sequence converges to 1.

Q2 Solution (B): Rewrite $a_n = \dfrac{2^n - 1}{2^n} = 1 - \left(\frac{1}{2}\right)^n$. As $n \to \infty$, $\left(\frac{1}{2}\right)^n \to 0$, so $\lim_{n\to\infty} a_n = 1 - 0 = 1$.

Q3 Solution (A): Split the fraction: $a_n = \left(\frac{2}{3}\right)^n - \left(\frac{1}{3}\right)^n$. Since $|2/3| < 1$ and $|1/3| < 1$, both terms approach 0. Thus, $\lim_{n\to\infty} a_n = 0$.

Q4 Solution (B): Let $x = 1/n$. As $n \to \infty$, $x \to 0^+$. The expression becomes $\lim_{x\to 0^+} \dfrac{\sin x}{x} = 1$.

Q5 Solution (A): Factor the denominator: $n^2 + 5n + 6 = (n+2)(n+3)$. For $n \geq 1$, $a_n = \dfrac{n+3}{(n+2)(n+3)} = \dfrac{1}{n+2}$. Taking the limit gives $\lim_{n\to\infty} \dfrac{1}{n+2} = 0$.

Q6 Solution (A): Applying L'Hôpital's Rule to $f(x) = \dfrac{x}{2^x}$ yields $\lim_{x\to\infty} \dfrac{1}{2^x \ln 2} = 0$. Thus $a_n \to 0$.

Q7 Solution (B): Invert the base: $\left(\dfrac{n}{n+1}\right)^n = \dfrac{1}{\left(\frac{n+1}{n}\right)^n} = \dfrac{1}{\left(1+\frac{1}{n}\right)^n}$. Since $\lim_{n\to\infty} \left(1+\dfrac{1}{n}\right)^n = e$, the limit is $\dfrac{1}{e}$.

Q8 Solution (A): Multiply numerator and denominator by the conjugate $\left(\sqrt{n^2-1} + \sqrt{n^2+n}\right)$:

$$a_n = \frac{\sqrt{n^2-1} + \sqrt{n^2+n}}{(n^2-1) - (n^2+n)} = \frac{\sqrt{n^2-1} + \sqrt{n^2+n}}{-n-1}$$

Divide numerator and denominator by n:

$$a_n = \frac{\sqrt{1-1/n^2} + \sqrt{1+1/n}}{-1-1/n} \longrightarrow \frac{1+1}{-1} = -2$$

Q9 Solution (B): Rationalize by multiplying by $\dfrac{n+\sqrt{n^2-n}}{n+\sqrt{n^2-n}}$:

$$a_n = \frac{n^2-(n^2-n)}{n+\sqrt{n^2-n}} = \frac{n}{n+\sqrt{n^2-n}} = \frac{1}{1+\sqrt{1-1/n}} \longrightarrow \frac{1}{1+1} = \frac{1}{2}$$

Q10 Solution (C): Rewrite $a_n = \dfrac{1}{(9/10)^n} = \left(\dfrac{10}{9}\right)^n$. Since $\dfrac{10}{9} > 1$, the terms grow without bound: $\lim\limits_{n\to\infty} a_n = +\infty$. The sequence diverges.